□

In many applications it suffices to use the following more convenient bound (see Corollary 2.3 in [8]).

Corollary 5. *Let X be distributed binomially with parameters n and p. If $0 < \varepsilon \leqslant \frac{3}{2}$, then*

$$\mathbb{P}\left(|X - np| \geqslant \varepsilon np\right) \leqslant 2e^{-\varepsilon^2 np/3}.$$

□

The following consequence of the Chernoff bounds can be obtained by applying Lemma 3 to X_i with $x_i = \frac{k}{n_i}$ (full details are given in [7]).

Corollary 6. *Let $(n_i)_{i\in\mathbb{N}}$ be a sequence of integers, and let $(p_i)_{i\in\mathbb{N}}$ be a sequence of probabilities. Let $X_i \sim \mathrm{Bin}(n_i, p_i)$, and let $k \in \mathbb{N}$ be constant. Suppose that $\mu_i := n_i p_i \to \infty$ as $i \to \infty$. Then*

$$\mathbb{P}(X_i \leqslant k) = O(\mu_i^k e^{-\mu_i}).$$

□

We will also need the following easy observation on the probability of a union of events.

Lemma 7. *Let A_i, $i = 1, \ldots, k$, be events in a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then*

$$\sum_i \mathbb{P}\left(A_i\right) - \sum_i \sum_{j<i} \mathbb{P}(A_i \cap A_j) \leqslant \mathbb{P}\left(\bigcup_i A_i\right) \leqslant \sum_i \mathbb{P}\left(A_i\right).$$

□

2.2 General observations

For the rest of the paper, define $p = p(n)$ as in Theorem 2, let $G \sim \mathcal{G}(n, p)$ and colour the edges of G independently and uniformly at random using r colours.

Lemma 8. *Let $\delta > 0$ be constant, let $W \subset V$ be a set of vertices with $|W| \sim n$, and let $v \in V$. Then for every colour, with probability $1 - o\left(\exp\left(-n^{1/r}\right)\right)$, there are at least $\frac{1-\delta}{r} np$ and at most $\frac{1+\delta}{r} np$ edges between v and W of the given colour.*

Proof. The number of such edges is distributed binomially with parameters $|W|$ (or $|W|-1$ if $v \in W$) and p/r. Since $|W|p/r \sim ((1+\varepsilon)Cn\log n)^{1/r}/r$, the probability that there are fewer than $\frac{1-\delta}{r}np$ or more than $\frac{1+\delta}{r}np$ such edges is $o\left(\exp\left(-n^{1/r}\right)\right)$ by Corollary 5. $\square$

For $k \in \mathbb{N}$, we call a path of length k in G a *k-path*, so a k-path is of the form $x_0x_1\dots x_k$ where the x_i are distinct vertices. We call a collection of paths in the graph *independent* if no two of them share any inner vertices.

Lemma 9. *There is a constant $c > 0$ such that whp every pair of vertices in G is joined by at least $c\log n$ independent r-paths.*

Proof. Fix two distinct vertices v and w. We shall explore the $(r-1)$-neighbourhood of v step-by-step, and apply Lemma 8 at each step with a suitable $\delta > 0$ to see that the sets we discover have the right size. Let $\delta > 0$ be such that $(1-\delta)(1+\varepsilon)^{1/r} = (1+\frac{\varepsilon}{2})^{1/r}$.

We start by considering all edges between v and $W_1 = V \setminus \{v, w\}$. By Lemma 8, with probability $1 - o\left(\exp\left(-n^{1/r}\right)\right)$ there are between $(1-\delta)np$ and $(1+\delta)np$ edges between v and W_1. Condition on this, and denote by N_1 the set of vertices in W_1 which are adjacent to v.

Next, let $W_2 = W_1 \setminus N_1$, and consider the edges between N_1 and W_2. Note that by our condition on the size of N_1, $|W_2| \sim n$. Furthermore, the edges between N_1 and W_2 are disjoint from and therefore independent of the edges we have considered so far. We go through the vertices z in N_1 one after the other, revealing the edges present. However, we disregard any edges to vertices which are adjacent to another vertex in N_1 which was considered earlier, so that the edges revealed form a tree. Applying Lemma 8 at each step, we can see that with probability $1 - |N_1|o\left(\exp\left(-n^{1/r}\right)\right) = 1 - o\left(\exp\left(-n^{1/2r}\right)\right)$, at each step there are between $(1-\delta)np$ and $(1+\delta)np$ edges.

Denote by N_2 the set of vertices in W_2 adjacent to a vertex in N_1, and let $W_3 = W_2 \setminus N_2$. We now proceed in the same way and explore the neighbours in W_3 of all vertices in N_2 disjointly, conditional on the neighbourhoods so far having the right sizes for all vertices according to Lemma 8.

We continue in this way until we have explored the entire $(r-1)$-neighbourhood N_{r-1} of the vertex v in W_1. Note that if all neighbourhoods have the right size, in total $O\left((np)^{r-1}\right) = o(n)$ vertices are revealed, so we can apply Lemma 8 at each step. With probability $1 - o(\exp\left(-n^{1/2r}\right))$, we now have a tree with at least $((1-\delta)np)^{r-1} = \left((1+\frac{\varepsilon}{2})Cn\log n\right)^{(r-1)/r}$ leaves. We can group the leaves together depending on which of the edges incident with v their path to v contains (i.e., which vertex in N_1 they originate

from) — each group has size at least $\left((1+\frac{\varepsilon}{2})Cn\log n\right)^{(r-2)/r}$, and there are at least $\left(\left(1+\frac{\varepsilon}{2}\right)Cn\log n\right)^{1/r}$ groups. The edges between w and the leaves of the tree are independent from the edges that have been explored before. The probability that w has a neighbour in a given vertex group is therefore at least

$$\begin{aligned}1-(1-p)^{\left((1+\frac{\varepsilon}{2})Cn\log n\right)^{(r-2)/r}} &\geqslant 1-\exp\left(-p\left(\left(1+\frac{\varepsilon}{2}\right)Cn\log n\right)^{(r-2)/r}\right)\\ &\geqslant 1-\exp\left(-\left(\left(1+\frac{\varepsilon}{2}\right)C\log n\right)^{(r-1)/r}n^{-1/r}\right)\\ &\sim \left(\left(1+\frac{\varepsilon}{2}\right)C\log n\right)^{(r-1)/r}n^{-1/r},\end{aligned}$$

using the fact that $1-x\leqslant\exp(-x)$ for all $x\in\mathbb{R}$ and that $1-\exp(-x)\sim x$ as $x\to 0$. These events are independent for the different vertex groups, so the number of groups with at least one edge to w is distributed binomially. If we pick one edge from each such vertex group, this gives independent paths from v to w by construction. Since there are at least $\left((1+\frac{\varepsilon}{2})Cn\log n\right)^{1/r}$ vertex groups, the expected number of such paths is at least $(1+\frac{\varepsilon}{3})C\log n$ if n is large enough. Note that $\varphi(x)\to 1$ as $x\searrow -1$, where φ is the function defined in Corollary 4. Therefore, since $r\geqslant 3$ and $C=\frac{r^{r-2}}{(r-2)!}>2$, if we pick $c>0$ small enough, the probability that there are fewer than $c\log n$ independent r-paths joining v and w is $o(n^{-2})$ by Corollary 4. □

2.3 The main lemma

Let

$$L=\left\lceil\frac{17r}{\varepsilon(r-1)}\right\rceil, K=rL, \text{ and } S=2Lr^2+2.$$

Call a pair of vertices *dangerous* if it is joined by at most K independent rainbow r-paths in the random colouring. The following lemma will form the main part of the proof.

Lemma 10. *For a pair $\{v,w\}$ of vertices, denote by $A_{v,w}$ the event shown in Figure 1: There are L independent r-paths $P_1,\dots,P_L$ joining v and w, and L r-paths $Q_1,\dots,Q_L$ such that, writing $\{x_i,y_i\}$ for the end vertices of Q_i, the following conditions hold:*

i) For each i, P_i contains an edge e_i that is also on Q_i.

ii) The pairs $\{x_i,y_i\}$, $i=1,\dots,L$, and $\{v,w\}$ are distinct.

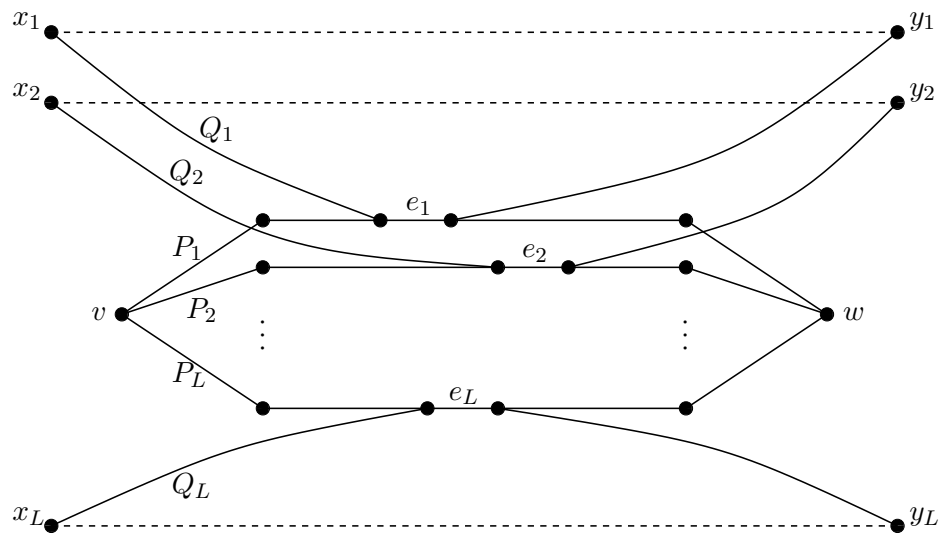

Figure 1: The event $A_{v,w}$. Dashed lines show dangerous pairs. The paths P_i only meet at v and w, while the paths Q_i may share vertices with each other and with the paths P_i. The pairs $\{x_i, y_i\}$ are distinct, but not necessarily disjoint.

iii) All pairs $\{x_i, y_i\}$, $i = 1, \ldots, L$, are dangerous.

Then whp $A_{v,w}$ does not hold for any pair $\{v, w\}$ of vertices.

The idea of the proof is the following. For one pair $\{x_i, y_i\}$ as in the lemma, the expected number of rainbow r-paths joining x_i and y_i is roughly $\frac{r!}{r^r} n^{r-1} p^r = \frac{r!}{r^r} C(1+\varepsilon)\log n = \frac{r-1}{r}(1+\varepsilon)\log n$. Since the rainbow paths behave roughly binomially, the probability that $\{x_i, y_i\}$ is dangerous is about $n^{-\frac{r-1}{r}(1+\varepsilon)}(\log n)^K$ by Corollary 6.

Therefore, given v and w, the probability that there is one path P_i containing an edge e_i which also lies on an r-path joining a dangerous pair $\{x_i, y_i\}$ is about

$$O^*\left(n^{2r-2} p^{2r-1} n^{-\frac{r-1}{r}(1+\varepsilon)}\right) = O^*(n^{-\varepsilon\frac{r-1}{r}}).$$

Therefore, if we can show that these events do not depend on each other too much for different P_i, then we would expect that the overall probability that there are L such paths is about $O^*\left(n^{-L\varepsilon\frac{r-1}{r}}\right)$. If L is chosen large enough, this will then be $o(n^{-2})$, completing the proof of the lemma.

A formal proof of this idea requires some care.